

Multiple Choice (10 marks)

1. Which mass is undergoing the greatest amount of acceleration?
 - (a) 1 kg subjected to a force of 1 N
 - (b) 1 kg subjected to a force of 100 N
 - (c) 100 kg subjected to a force of 1 N
 - (d) 100 kg subjected to a force of 100 N
2. Rocks sometimes show evidence of ancient glaciers. Which is the best sign that glaciers once covered an area?
 - (a) Rocks have many scratches.
 - (b) Rocks contain fossils.
 - (c) Rocks are covered by moss.
 - (d) Rocks are deep underground.
3. Planets orbit
 - (a) Earth.
 - (b) the Moon.
 - (c) the galaxy.
 - (d) the Sun.
4. Which group consists only of living things?
 - (a) rabbit, seed, bird
 - (b) seed, bird, wind
 - (c) Volcano, candle, rabbit
 - (d) wind, candle, volcano
5. An engineer is analyzing which areas in a city might become flooded if there are heavy rains. Which of the following maps is best to use for this analysis?
 - (a) a map showing the routes of city buses
 - (b) a map showing the locations of streets
 - (c) a map showing the locations of houses
 - (d) a map showing the elevations of ground surfaces

True / False (10 marks)

Answer True or False

6. True or False: A black hole occupies more space than a galaxy due to its immense gravitational pull.
7. True or False: Preventing sperm production in male insects can help reduce the total population of insects.
8. True or False: Pioneer plants must have the ability to grow very large to succeed on recovering beaches.
9. True or False: Reef ecosystems require organisms to cope with daily dry periods just like intertidal ecosystems do.
10. True or False: Enzymes enable biochemical reactions to occur at lower temperatures than would otherwise be possible.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the steps Pat should take to determine which plant food, "Quickgrow" or "Supergrow," is more effective for the growth of a specific houseplant, including how to measure growth.
12. Discuss a specific property of matter and explain how it distinguishes different types of matter, providing examples to support your analysis.

End of Paper